CHATGPT-LIKE AI DEVELOPMENT

20-step roadmap • Beginner to advanced • Python-focused learning path

Goal: Learn the skills needed to build an AI assistant using existing models first, then progress toward training and deploying your own language models.

PHASE 1 — Programming & Mathematics

Step	What to learn	Focus
1	**Python Programming**	Variables, data types, conditions, loops, functions, modules and error handling.
2	**Advanced Python**	OOP, decorators, generators, iterators, typing, async programming and project structure.
3	**Data Structures & Algorithms**	Arrays, strings, hash maps, stacks, queues, trees, graphs, sorting and complexity.
4	**NumPy**	Arrays, vectorization, broadcasting and numerical operations.
5	**Pandas**	Data cleaning, transformation, analysis and dataset preparation.
6	**Linear Algebra**	Vectors, matrices, dot products, eigenvalues and matrix operations.
7	**Calculus**	Derivatives, gradients, chain rule and optimization foundations.
8	**Probability & Statistics**	Distributions, expectation, variance, sampling and evaluation concepts.

PHASE 2 — Artificial Intelligence

Step	What to learn	Focus
9	**Machine Learning**	Supervised and unsupervised learning, model evaluation and Scikit-learn.
10	**Deep Learning**	Training neural networks, loss functions, optimizers and regularization.
11	**Neural Networks**	Layers, activations, forward propagation and backpropagation.
12	**PyTorch**	Tensors, autograd, datasets, data loaders, model building and training loops.
13	**GPU Computing**	GPU fundamentals, parallel computation and CUDA concepts.

PHASE 3 — Language Models & ChatGPT Architecture

Step	What to learn	Focus
14	**Natural Language Processing (NLP)**	Text preprocessing, language representation and language-model basics.
15	**Tokenization & Embeddings**	Subword tokenization, vocabulary, token IDs and vector representations.
16	**Attention Mechanism**	Queries, keys, values, self-attention and multi-head attention.
17	**Transformer Architecture**	Positional information, encoder/decoder concepts and transformer blocks.
18	**Large Language Models (LLMs)**	Pretrained models, context windows, prompting and model families.

PHASE 4 — Training & Product Development

Step	What to learn	Focus
19	**Model Training & Evaluation**	Pretraining concepts, fine-tuning, instruction tuning, preference optimization/RLHF concepts, safety and evaluation.
20	**Inference, Deployment & Applications**	Model serving, APIs, backend, databases, monitoring, scaling and building a chat interface.

RECOMMENDED LEARNING ORDER

1	Python + Mathematics
2	Machine Learning + Scikit-learn
3	Deep Learning + PyTorch
4	NLP + Tokenization + Attention
5	Transformers + LLMs
6	Fine-tuning + Evaluation
7	Inference + Backend + Deployment

Important distinction

Building an AI assistant by connecting or fine-tuning an existing open-source model is much more achievable as a first project. Training a frontier-scale model from scratch requires major research expertise, large datasets, specialized hardware, engineering teams and substantial infrastructure. This roadmap teaches the underlying skills; it is not a guarantee that completing 20 steps alone will reproduce ChatGPT.

Suggested first project: Build a simple Python chatbot using an existing language-model API, then build a small retrieval-augmented (RAG) assistant over your own notes. Later, experiment with fine-tuning a small open model.